

Faithfulness Benchmarking for Bangla RAG

A comparative analysis of seven RAG evaluation frameworks, and a research proposal for end-to-end Bangla RAG faithfulness evaluation

3.4 Benchmark by benchmark

The matrices above compress; this section restates each framework in full against the eleven review questions, with the exact figures a reader would want to quote.

3.4.1 RAGTruth (ACL 2024)

- What is benchmarked. Two objects at once: the hallucination frequency and density of six LLMs under RAG conditions, and the accuracy of four families of hallucination detector measured against human labels. A third experiment uses the detector to suppress hallucination by choosing between candidate responses.
- Research objective. To supply what the field lacked ; a large-scale, high-quality corpus of *naturally occurring* RAG hallucinations annotated at word level, rather than synthetically injected errors.
- Dataset and language. English only. 2,965 instances spanning QA (989, from MS MARCO), data-to-text (1,033, from Yelp business records) and summarisation (943, from CNN/Daily Mail plus recent news); six responses per instance gives 17,790 responses, of which 7,664 (43.1%) contain hallucination, totalling 14,289 annotated spans. The test split is 450 instances, 150 per task.
- Data generation and annotation. Responses generated naturally by GPT-3.5-turbo-0613, GPT-4-0613, Mistral-7B-Instruct and Llama-2-chat at 7B, 13B and 70B, using plain task prompts. Annotation was performed by a professional vendors degree-qualified annotators in Label Studio, two independently per response with third-pass adjudication. The taxonomy has four classes ; Evident Conflict, Subtle Conflict, Evident Introduction of Baseless Information, Subtle Introduction of Baseless Information ; plus two adaptive tags, `implicit_true` (13.5% of spans) and `due_to_null` (11.5%).
- Dimensions and metrics. For generators: counts of hallucinated responses and hallucination density in spans per 100 words. For detectors: response-level precision, recall and F1, and character-level span precision, recall and F1.
- Human evaluation. Yes, and it is the foundation of the resource. Reported agreement is 91.8% at response level and 78.8% at span level; the paper does not report Cohens or Fleiss kappa, which is its clearest methodological omission.
- LLM as judge. Only as a baseline detector ; prompted GPT-4-turbo and GPT-3.5-turbo emitting JSON span lists, plus SelfCheckGPT and LMvLM.
- Baselines. Prompted GPT-3.5-turbo 52.9 F1, prompted GPT-4-turbo 63.4, SelfCheckGPT 58.8, LMvLM 49.4 ; all beaten by a fine-tuned Llama-2-13B at 78.7 response-level and 52.7 span-level F1. Response selection using the detector cuts hallucinated responses by up to 63.2%.
- Reproducibility. The corpus is public on GitHub, generation and detection prompts are printed in full, and model versions are pinned. Detector weights are not released and no licence is stated.
- Major limitations. Manual annotation is expensive and does not scale ; the paper notes the cost (\$25 per hour, two annotators per response) but raises it in the introduction, not as a stated limitation. English only, which the paper never flags at all. Generators are 2023-era, so the error distribution may already be dated. One caveat on attribution: it is true that the corpus measures generation rather than retrieval, since contexts are taken from datasets rather than produced by a live retriever, but the paper never frames this as a limitation and it should not be quoted as the authors' own view.
- Gap left open. Whether detectors trained on this corpus generalise to other domains ; the authors' own caveat ; whether manual and synthetic annotation can be productively combined, and the whole of the non-English world.

3.4.2 RAGAS (EACL 2024)

- What is benchmarked. Complete RAG pipelines, decomposed into the retrievers focus, the generators grounding in retrieved passages, and the answers relevance to the question.
- Research objective. Reference-free evaluation: scores computed without any gold answer, so that teams can iterate on a pipeline without commissioning annotation.

- Dataset and language. English only. WikiEval comprises 50 question–context–answer items drawn from Wikipedia pages on post-2022 events, chosen to fall outside the judge models training window.
- Data generation and annotation. ChatGPT generates a question and answer per page, then deliberately degraded variants: an answer produced with no context at all (low faithfulness), an answer instructed to be incomplete (low answer relevance), and a context padded with back-link material (low context relevance). Two annotators expressed pairwise preferences, agreeing about 95% of the time on faithfulness and context relevance and 90% on answer relevance.
- Dimensions and metrics. Faithfulness is the fraction of decomposed statements an LLM judges supported by the context. Answer relevance is the mean cosine similarity between the original question and questions regenerated from the answer. Context relevance is the proportion of context sentences the LLM marks as essential.
- Human evaluation. Present but minimal, and used only to validate the metrics rather than to score systems. On the 50 pairs, RAGAS agrees with human preference 0.95 / 0.78 / 0.70 across the three dimensions. *(The pasted analysis is internally inconsistent here ; one line says human evaluation is involved, the table says "No". Both are defensible shorthand for the same fact: humans validate the metric, they never label the systems under test.)*
- LLM as judge. Yes, throughout: gpt-3.5-turbo-16k computes all three metrics, with text-embedding-ada-002 supplying the answer-relevance embeddings.
- Baselines. GPT Score, which asks ChatGPT for a 0–10 rating, scores 0.72 / 0.52 / 0.63; GPT Ranking, which asks for a direct pairwise choice, scores 0.54 / 0.40 / 0.52.
- Reproducibility. The ragas library is open-source with LangChain and LlamaIndex integrations, and requesting JSON-formatted judge output measurably stabilises scores across runs. The deeper problem is unaddressed: every number depends on a closed model that changes underneath the benchmark.
- Major limitations. A 50-item validation set with two annotators and no significance testing; synthetic degradations that are easier to detect than real hallucinations; scalar outputs with no indication of *where* an answer went wrong.
- Gap left open. Rigorous large-scale meta-evaluation, which ARES and MEMERAG later supplied; behaviour on long contexts and subtle nuance; and whether any step of the pipeline survives translation to a morphologically rich, lower-resource language.

3.4.3 ARES (NAACL 2024)

- What is benchmarked. RAG systems are ranked both mock systems with known accuracy between 70% and 90% and real retriever × generator combinations and ARES itself is validated by how closely its rankings match ground truth. One point of terminology worth correcting, because it recurs in summaries of this literature: ARES is not an *end-to-end* framework. It scores the three components separately with three separate judges. MEMERAG is the end-to-end one that is what the "E" in its name denotes.
- Research objective. To make automated evaluation trustworthy rather than merely cheap, by fine-tuning small judges on synthetic in-domain data and then correcting their errors statistically.
- Dataset and language. English throughout: KILT (Natural Questions, HotpotQA, FEVER, Wizard of Wikipedia) and SuperGLUE (MultiRC, ReCoRD), plus the AIS attribution sets. Cross-domain stress tests use XGLUE, CodeSearchNet and T-REx.
- Data generation and annotation. FLAN-T5 XXL generates synthetic query–answer pairs from in-domain passages, filtered by requiring the query to retrieve its own source passage. Negatives are constructed both weakly (random passages) and strongly (same-document passages, and prompted contradictory answers). A human preference set of roughly 150–300 labelled examples is the only manual input.

- Dimensions and metrics. Three DeBERTa-v3-Large binary judges, one per dimension of the RAG triad; prediction-powered inference rectifies their errors and produces 95% confidence intervals; system rankings are scored by Kendalls τ , with τ above 0.9 treated as success.
- Human evaluation. Yes, but deliberately small and the paper quantifies how small it can be: below roughly 100–150 labels, ARES cannot reliably distinguish systems. It beats a sampled-annotation baseline that uses 1,350 labels while consuming 78% fewer.
- LLM as judge. LLMs generate the training data; a few-shot GPT-3.5 judge serves as a baseline; GPT-4 was tested as a substitute for human annotators and found to cost 0.05–0.30 of Kendalls τ .
- Baselines. RAGAS v0.0.18, beaten on every dataset with judge-accuracy gaps up to 75.3 points; a few-shot GPT-3.5 judge, beaten by 0.06 τ on average; and sampled human annotation.
- Reproducibility. Code and datasets are public, all prompts and hyperparameters are in the appendices, and porting to a new domain requires only in-domain passages, about 150 human labels and five few-shot examples.
- Major limitations. It needs GPUs and domain-literate annotators. More importantly for us, its judges degrade badly under distribution shift, and the sharpest case is language: on XGLUE, Kendalls τ falls to 0.33. The faithfulness judge is also the weakest of the three, reaching only 62.5% accuracy on the Wizard-of-Wikipedia attribution set.
- Gap left open. Multilingual judges, which the authors explicitly defer; span-level localisation, since each judge emits a single binary verdict; and any evidence at all for low-resource languages.

3.4.4 MEMERAG (ACL 2025)

- What is benchmarked. Automatic evaluators LLM judges under several prompting strategies meta-evaluated by how well they agree with expert human judgements of natively generated RAG answers.
- Research objective. To provide the multilingual meta-evaluation benchmark the field lacked, while avoiding translationese: the authors argue that translated benchmarks are distributionally unlike real text and therefore mismeasure both generators and judges.
- Dataset and language. Five languages English, German, Spanish, French and Hindi. Bengali is not among them. The benchmark is built on MIRACL, using 250 questions per language for 1,250 questions and 2,322 annotated sentences, with an extension set of 750 questions annotated five ways. Answers come from Claude 3 Sonnet, Llama 3 70B and 8B, Mistral 7B and GPT-4o mini.
- Data generation and annotation. Contexts are the top five BM25 passages, with at least one known-relevant passage guaranteed. Answers are generated natively at temperature 0.1. Professional native annotators label every sentence for faithfulness three coarse classes and ten fine-grained ones, including Adds new information, Contradiction, Mis-referencing, Nuance shift and Opinion as fact and for relevance across three classes.
- Dimensions and metrics. Judges are scored by balanced accuracy against the human labels, with permutation tests at $\alpha = 0.05$ and bootstrap standard errors.
- Human evaluation. Yes, and rigorously: a flow-chart-guided interface raised inter-annotator agreement from Gwets AC1 0.45 in piloting to 0.84–0.93, with Fleiss K 0.70–0.88. Fine-grained agreement remains much lower (AC1 0.18–0.63), which is itself a useful warning.
- LLM as judge. The judges are the object of study: GPT-4o mini, Qwen 2.5 32B and Llama 3.2 at 90B and 11B, each under zero-shot, chain-of-thought, annotation-guidelines and AG+CoT prompting. Guidelines-in-prompt is consistently the largest single lever.
- Baselines. The judge $\times$ prompt grid itself; RAGAS and ARES are cited but not run. Balanced accuracy spans 55.4 to 76.8, with the best multilingual configuration at 72.6.
- Reproducibility. The benchmark, the extension set, the annotation flow chart and all prompts are released on GitHub. No licence is stated.

- Major limitations. Five languages, all Indo-European; questions are not parallel across languages, so cross-language comparison is confounded; 250 questions per language is thin once results are sliced by error type; annotation and judging are sentence-level, so nothing localises to a span.
- Gap left open. Every Indic language except Hindi, Bengali included; span-level localisation; fine-tuned multilingual evaluators, which the authors defer to future work; and the roughly 25-point headroom between the best judge and human agreement.

3.4.5 FaithJudge (EMNLP 2025, Industry Track)

- What is benchmarked. Generator faithfulness across summarisation, QA and data-to-text ; twelve LLMs in the new leaderboard, over 160 in its predecessor and, in the same experiments, the quality of the judges and detectors doing the measuring.
- Research objective. To lift automated faithfulness evaluation above the plateau where detectors sit near chance, without training anything: the judge is shown human-annotated hallucinations produced by *other* models on the *same* source document, and asked to assess a fresh response against that backdrop.
- Dataset and language. English. FaithBench supplies 75 articles with ten annotated summaries each; RAGTruth supplies up to 150 sources per task; AggreFact-SOTA and TofuEval-MeetingBank are used in the detector comparison.
- Data generation and annotation. No new annotation. FaithBenchs taxonomy Consistent, Benign, Questionable, Unwanted comes with spans and source references from multiple annotators, and its articles were selected adversarially where existing detectors disagreed. Meta-evaluation collapses this to Unwanted versus Consistent, since Benign and Questionable proved unreliable.
- Dimensions and metrics. Judge quality as balanced accuracy and macro-F1 against human labels; generator quality as hallucination rate on the leaderboard.
- Human evaluation. Yes, but amortised: humans built the example pool once, and every subsequent model is scored against it automatically. This is the economic insight of the paper.
- LLM as judge. The core mechanism. Qwen-2.5 at 7B and 72B, Llama-3.1-8B, Llama-3.3-70B, GPT-4o and o3-mini-high were all tested; o3-mini-high was adopted. A three-judge majority vote did not beat the best single judge, and judges were found to under-predict hallucination and to show measurable self-preference.
- Baselines. Fine-tuned detectors HHEM-1.0 and 2.1-open, AlignScore, MiniCheck, Bespoke-MiniCheck-7B and TrueTeacher; zero-shot prompting via RAGAS-style claim verification, FACTS Grounding and chain-of-thought. The best pre-FaithJudge result on FaithBench was 68.8 balanced accuracy; FaithJudge reaches 84.0.
- Reproducibility. Code and an evolving leaderboard are public; HHEM-2.1-open is on Hugging Face with over four million downloads; HHEMs own training details remain proprietary.
- Major limitations. The method can only judge responses to documents that already carry multi-model human annotation, which bounds coverage to 72–150 sources per task. It depends on a proprietary reasoning model whose cost, availability and version may drift. Verdicts are binary. And FaithBenchs adversarial selection makes absolute rates unrepresentative of typical inputs.
- Gap left open. Multilingual judging is entirely unaddressed, and the method presupposes an annotated pool that exists in no low-resource language; there is no distillation into a cheap local detector; and verdicts never localise to a span.

3.4.6 LettuceDetect and Turk-LettuceDetect (2025)

These are two papers, and the comparison matrix merges them for brevity but the merge hides the single most important difference between them, so they are separated here.

- What is benchmarked. Token-level hallucination detection: each token of a response is classified as supported or unsupported given the question and retrieved context, and results are reported at both example and span level.
- Research objective. LettuceDetect targets two weaknesses of the field at once the 512-token ceiling of classic encoders, and the latency and cost of LLM judges. Turk-LettuceDetect asks whether that recipe transfers to a morphologically complex, lower-resourced language.
- Dataset and language. LettuceDetect trains and evaluates on RAGTruth in English, where inputs average 801 tokens and reach 2,632 the empirical case for long context. Turk-LettuceDetect machine-translates the whole of RAGTruth into Turkish using Gemma-3-27B-IT, about twelve GPU-hours for a full pass, preserving hallucination spans by wrapping them in inline <HAL> tags before translation.
- Data generation and annotation. Neither paper performs new human annotation. LettuceDetect derives token labels from RAGTruths spans, masking context and question tokens from the loss. Turk-LettuceDetect projects those same labels through machine translation and its own worked example shows three English spans becoming two Turkish ones, an error rate it never measures.
- Dimensions and metrics. Precision, recall, macro-F1 and AUROC at token and example level, reported per task.
- Human evaluation. None in either paper. Turk-LettuceDetect has no human involvement at any stage no native annotation, no translation audit, no human check on the projected spans.
- LLM as judge. LettuceDetect uses none, comparing against prompted GPT-3.5 and GPT-4-turbo, SelfCheckGPT, LMvLM, ChainPoll, RAGAS and Trulens as baselines. Turk-LettuceDetect does run five LLMs (GPT-4.1, Qwen3-14B, QwenQ-32B, Mistral Small 3.1 and 3.2) as comparison detectors, finding they achieve very high recall up to 0.9938 at poor precision, over-flagging heavily.
- Baselines and results. LettuceDetect-large reaches 79.22 example-level F1 and 58.93 span-level F1, beating Luna (65.4), prompted GPT-4-turbo (63.4) and the fine-tuned Llama-2-13B (78.7), and losing only to RAG-HAT (83.9) at roughly thirty times the size. Turk-LettuceDetects best Turkish model reaches 0.7266 token-level F1, against 0.7922 for the English original; the three encoders trained were a Turkish ModernBERT (135M), a gte-multilingual-based model (210M) and EuroBERT-210m none of which covers Bengali.
- Reproducibility. LettuceDetect is exemplary: MIT-licensed code, a pip package and both models on Hugging Face. Turk-LettuceDetect claims an open release but names no repository or licence anywhere in the paper, and its conclusion says only that release is planned. It also contains several citation errors and a results table in which the AUROC column duplicates the recall column.
- Major limitations. Both are bound to a single benchmark with no out-of-distribution evidence, and both collapse RAGTruths typed taxonomy into a binary decision. Summarisation is markedly the weakest task in both. The decisive limitation is Turk-LettuceDetects: it trains *and evaluates* on translated text, so its numbers describe performance on translationese, not on Turkish as a native speaker produces it.
- Gap left open. LettuceDetect names no multilingual long-context encoder to substitute for ModernBERT. Turk-LettuceDetect leaves open everything native ; native generation, native annotation, translation-quality auditing, meta-evaluation against human judgement, and evaluation on a deployed system.

3.4.7 BenHalluEval (2026)

- What is benchmarked. The ability of nine LLMs to *detect* hallucination: each model gives a binary verdict on whether a candidate response is hallucinated with respect to a supplied context, under a dual-track protocol in which Track A presents 1,000 gold answers per task that should be accepted and Track B presents synthetic hallucinated candidates that should be rejected. Chain-of-thought prompting is benchmarked as the sole mitigation.

- Research objective. To establish the first systematic hallucination benchmark for Bengali, covering both native Bengali script and Romanised Bangla-English code-mixed input, using a calibration-aware metric that a model answering uniformly cannot game.
- Dataset and language. Bengali and Bangla-English code-mixed. Four tasks: QA seeded from TyDiQA-GoldP Bengali, code-mixed QA converted from the same seeds, summarisation from BanglaCHQ-Summ consumer-health data, and mathematical reasoning from SOMADHAN. In total 16,000 items ; 12,000 hallucinated candidates and 4,000 gold answers.
- Data generation and annotation. All 12,000 candidates were written by GPT-5.4 as single-fault perturbations, one hallucination type per prompt, across a twelve-category taxonomy adapted from HaluEval. Labels are therefore single-label by construction and response-level only. Three native Bengali annotators validated a stratified sample of about 1,000 items, reaching Fleiss K of 0.911 to 0.926 ; genuinely strong agreement.
- Dimensions and metrics. BenHalluScore, the mean of the false-alarm rate on gold answers and the miss rate on hallucinated ones ; a balanced error rate in which 0% is perfect and 50% is what a model answering at random or uniformly achieves.
- Human evaluation. Present at construction and validation time only ; label validation, plus a check that code-mixed conversion preserved meaning (κ 0.906) and read naturally (4.38 out of 5). Scoring at evaluation time is fully automatic.
- LLM as judge. Pervasive, and in an unusual configuration: the evaluated models *are* the judges. Seed filtering also used Qwen2.5-32B as a judge, validated against another LLM ; GPT-4.1 mini, agreeing 91–95% ; rather than against people.
- Baselines. Nine models across three categories: reasoning (DeepSeek-R1-14B, GPT-4.1 mini), multilingual (Qwen2.5-32B, Gemma-2-27B, Mistral-nemo-12B, Llama-3.1-8B) and Bangla-centric (TigerLLM-9B, TituLLM-3B, BanglaLLaMA-13B). GPT-4o, Gemini and Claude are absent.
- Headline findings. BenHalluScore spans 7.72% to 55.42%. Rankings are task-conditional rather than ordered by scale, and parameter count predicts calibration poorly. Bengali-centric pretraining is "neither necessary nor sufficient": TituLLM and BanglaLLaMA collapse into near-uniform verdicts, with TituLLM answering "No" on 99–100% of gold items. Chain-of-thought lowers error in 14 of 32 evaluable model–task combinations and *raises* it in 18. Most interesting for our purposes, Romanised code-mixing *improves* three multilingual models by up to 22.85 BenHalluScore points, which the authors attribute to script anchoring toward stronger English representations.
- Reproducibility. Data and code sit on an anonymous 4open.science repository; no licence is stated; two models were evaluated on only a 10% stratified sample; there is no leaderboard and no versioning scheme.
- Major limitations, in the authors' own words. The candidates are GPT-generated and so "may still reflect generation-specific patterns that differ from naturally occurring hallucinations in deployed systems" ; their first stated limitation. Equal weighting of the two tracks may not suit deployments with asymmetric costs. All evaluation is zero-shot at temperature zero. Generalisation to other Bengali genres and domains remains unvalidated.
- Gap left open. The whole of the RAG pipeline, which Section 4 examines in detail.

4. Validating the Bangla RAG faithfulness opportunity

4.1 What BenHalluEval established

BenHalluEval is a genuine and carefully executed contribution, and the proposal must begin by acknowledging exactly what it settled. It established that Bengali has a systematic hallucination benchmark; that a calibration-aware metric can stop a model gaming the task by answering uniformly; that native Bengali annotators can validate such a resource to Fleiss K between 0.911 and 0.926; and that

several findings about Bengali LLM behaviour are robust enough to build on. Three matter directly for our design:

- Bengali-centric pretraining is neither necessary nor sufficient. TigerLLM performs respectably on QA, while TituLLM and BanglaLLaMA collapse into near-uniform verdicts ; TituLLM answers "No" to 99–100% of gold items and BanglaLLaMA could not be evaluated under chain-of-thought at all because of repetition collapse. Any Bangla protocol must measure and report verdict-format compliance, and must not select generators for Bangla branding over generation competence.
- Chain-of-thought is not a reliable mitigation in Bengali. It reduced error in 14 of 32 model–task combinations and increased it in 18. Prompting alone is not the answer, which is itself an argument for grounding.
- Script matters. Romanised code-mixing *improved* several multilingual models ; Qwen2.5-32B moves from 46.85 to 24.00 BenHalluScore between native and code-mixed QA. The authors attribute this to Roman script anchoring the model toward stronger English representations.

4.2 Detection versus end-to-end evaluation

What separates BenHalluEval from a RAG benchmark is not quality but object. BenHalluEval measures whether a model can detect that a supplied answer is hallucinated. RAG evaluation in the RAGAS and ARES lineage measures something strictly larger: whether a retriever surfaces relevant documents, and whether a generator then uses those specific documents faithfully.

The two come apart in a way that matters for deployment. A model can be excellent at spotting a planted error in someone elses answer and still hallucinate freely when writing its own, and no published result establishes the correlation for Bengali. More importantly, a detection benchmark cannot attribute failure: when a deployed Bangla RAG system produces an unsupported claim, the engineer needs to know whether the retriever failed to surface the evidence or the generator ignored evidence it had, and BenHalluEvals design cannot answer that even in principle, because retrieval never happens.

4.3 Static context versus dynamic retrieval

BenHalluEval uses static question–passage pairs inherited from TyDiQA-GoldP and BanglaCHQ-Summ. The supplied passage is the gold passage: by construction it contains the answer. Real retrieval does not behave this way ; relevance is partial and graded, passages may be topically related but unhelpful or mutually contradictory, and often enough to matter nothing relevant is returned at all.

For Bengali this is not a hypothetical concern. Recent work on Bangla agricultural RAG measured dense retrieval recall@10 swinging from 0.093 on colloquial farmer queries to 0.970 on formal safety queries, and BM25 recall@10 collapsing from 0.506 to 0.004 under cross-lingual query–corpus mismatch. Retrieval-induced hallucination is a first-order, quantitatively large problem in Bangla specifically, and no gold-passage benchmark can surface it.

4.4 What already exists ; the honest landscape

This is the section that changed most between drafts, and it must appear early in any paper we write. Eight adjacent works have a claim on this territory, and each must be cited and differentiated precisely.

Existing work	What it has for Bengali	What it does not do
BenHalluEval (arXiv:2605.31483)	16,000 items; claims "first dedicated hallucination benchmark for Bengali"; Fleiss K 0.911–0.926	No retrieval anywhere in 25 pages; GPT-5.4-synthesised errors; ~1,000 of 16,000 items human-validated; response-level binary only; no span localisation; no human-versus-constructed-label concordance
MultiWikiQHALLuA (arXiv:2605.02504)	A Bengali split on Hugging Face: 7,990 context-grounded,	Fully synthetic LettuceDetect-style injection; zero human validation in any language; gold Wikipedia context, not retrieval; the paper reports no Bengali results at all, and its classifiers cover only 30 European languages

Existing work	What it has for Bengali	What it does not do
	span-labelled rows with hallucinated-parts fields	
SHROOM-CAP (CHOMPS 2025 @ AACL)	Bengali present among 9 languages	Zero-shot test-only ; no in-language training data or guidelines; binary factual and fluency labels, not spans; scientific-publication metadata, not a RAG pipeline
MIRAGE-Bench (NAACL 2025)	Bengali among 18 languages, ~500 queries	Oracle passages by design ; retrieval deliberately factored out; arena-style rankings from a surrogate judge distilled from GPT-4o; no per-instance human faithfulness labels
NoMIRACL (Findings of EMNLP 2024)	Bengali present, with 1,125 human-judged all-non-relevant and 1,541 relevant queries; defines hallucination rate and error rate over retrieval subsets	A binary probe ; the model answers "Yes, answer is present" or "I don't know". No free-text scoring, no spans, no taxonomy
MEMERAG (ACL 2025)	The method we would be extending	Five languages, Hindi the only Indic one; Bengali never mentioned; retrieval held fixed at five passages; sentence-level, and its fine-grained agreement collapsed to AC1 0.18–0.63
Facet-Level Tracing (arXiv:2604.09174)	;	Already separates retrieval-induced from generator-intrinsic hallucination via Strict RAG / Soft RAG / LLM-only modes with a five-way taxonomy. English-only, and releases no corpus
BHRAM-IL (arXiv:2512.01852)	;	The flagship Indic hallucination benchmark: Hindi, Gujarati, Marathi, Odia. Bengali conspicuously absent

Two further items must be checked manually before any submission: a live Kaggle competition, "অলীকবচন ; Bengali LLM Hallucination Detection Challenge", whose dataset format and status could not be extracted programmatically; and the exact generation metrics used by IndicRAGSuite, which covers thirteen Indian languages including Bengali.

4.5 The gap, stated precisely

After all of that, five statements remain true, and together they are the proposals justification.

1. No Bengali resource contains naturally generated model output that has been human-annotated for faithfulness. Everything that exists is either synthetic (BenHalluEval, MultiWikiQHALLUA) or annotates the knowledge base rather than the models responses (KrishokChat validates its knowledge nodes at Cohens κ 0.82, not its generations).
2. No Bengali resource pairs real retrieved evidence with span-level faithfulness labels. MIRAGE-Bench and MEMERAG use oracle or fixed passages; NoMIRACL has real retrieval but only a binary probe; BenHalluEval has neither.
3. Bangla is the only major world language without a RAGTruth-style port, and the structural reason is checkable: EuroBERTs fifteen languages stop at Hindi.
4. No one has measured how far an LLM judge can be trusted on Bengali faithfulness. Given that judges flip 10.7–14.4% of pairwise preferences under language-switched presentation, that multilingual judge agreement averages Fleiss κ around 0.3 across 25 languages, and that ARESs judges fall to τ 0.33 under language shift, the honest prior is that they cannot be trusted ; and that is worth establishing rather than assuming.
5. The separation of retrieval-induced from generator-intrinsic hallucination has been done as a diagnostic method, in English, with no released corpus. Nobody has operationalised it as ground truth baked into data collection, in any language.

The encouraging half is that every ingredient now exists. MIRACL and NoMIRACL provide Apache-2.0 Bengali passages with human relevance judgements. mmBERT provides a multilingual encoder in which

Bengali enters at mid-training with 3.9 billion tokens at the full 8,192-token context, and which PsiloQA has already shown to be the best available encoder for multilingual span-level hallucination detection. BanglaQuAD and BanglaRQA provide native-authored Bengali questions that BenHalluEval has not consumed. This is an integration and annotation gap, not a technology gap.

5. Proposed research directions

Three directions follow. They are composable, and adversarial review was clear that they should not all be attempted at full depth in one paper.

5.1 Direction 1 ; Bangla-RAG: an end-to-end pipeline benchmark with attribution

Context. BenHalluEval benchmarks whether LLMs can detect a hallucination given a static question and a gold passage. It does not evaluate the retrieval-augmented half of the pipeline at all, and the retrieved context is itself a cause of hallucination that no gold-passage benchmark can observe.

Research question. How does evidence quality shape faithfulness in Bangla RAG, and how much of the observed unfaithfulness is retrieval-induced rather than generator-intrinsic ; measured by construction rather than inferred by a judge?

How it must differ from what exists. Two things, and they should be the paper. First, evidence quality becomes a controlled independent variable: using MIRACL Bengalis per-passage human judgements, contexts are built with k of 5 passages relevant for k in $\{5, 3, 1, 0\}$, with NoMIRACLs human-verified all-non-relevant Bengali queries supplying the $k = 0$ arm. That produces a dose-response faithfulness curve that neither NoMIRACLs two-bucket design nor MEMERAGs fixed five passages can produce, for any language. Second, span-level Bengali annotation that survives the agreement cliff ; MEMERAGs fine-grained agreement fell to Gwets AC1 0.18 in Hindi, which is why its judge experiments use coarse labels only. Getting span-level Bangla annotation to a defensible floor is a methodological contribution in itself.

Contribution. Moves Bengali evaluation from a static detection task to a pipeline task with ground-truth attribution, and releases the corpus that the English-only diagnostic literature never produced.

5.2 Direction 2 ; Cross-lingual transfer and the value of native supervision

Context. Turk-LettuceDetect showed that machine-translating RAGTruth and fine-tuning a compact encoder reaches 0.7266 token-level F1 in Turkish. What it could not show, having no native data, is whether those numbers hold on text a native speaker actually produced. Meanwhile a January 2026 study found that hallucination detectors are robust within a language but fail cross-lingually without in-language supervision ; while TrueTeacher reports a Bengali mFACE ROC-AUC of 98.5 against a 45-language average of 77.2, which points the other way. The evidence is genuinely contested.

Research question. How much of the cross-lingual evaluator gap in Bangla is attributable to translationese in training, translationese in test, domain and generator shift, and genuinely Bengali-specific error types ; and how many native annotated examples does it take to close it?

The single highest-value experiment is a native-data scaling curve: train at 0, 100, 250, 500, 1,000 and all native Bengali annotated responses on top of English-only and machine-translated initialisation, and report the curve. That converts a yes-or-no question into an actionable number for every low-resource language, and it publishes whichever way the curve bends. The expected regime is set by BLUB: zero-shot transfer into Bengali costs about 6 points on sentence classification but about 17.5 points overall, because span and sequence labelling transfer far worse.

Contribution. A cost-efficiency analysis practitioners can act on, plus the first three-way comparison of training-data provenance ; English-only, machine-translated, and unvalidated synthetic Bengali ; against native supervision.

5.3 Direction 3 ; Script anchoring in multilingual RAG evaluation

Context. BenHalluEval found several multilingual models did better on Romanised code-mixed input than on native Bengali script. Whether that extends to generation under retrieval, and whether it biases the *judges* used to score that generation, is unknown.

Research question. Are RAG faithfulness evaluators script-invariant, and is the non-invariance large enough to reorder a generator leaderboard?

The honest positioning. Parts of this are already published and must not be claimed: that judges are not invariant to the surface language of what they read is established, with 10.7–14.4% preference flips across 13,408 judgments. The unoccupied territory is the interaction ; script-of-evidence crossed with script-of-response, on semantically certified counterfactual pairs, with sensitivity and decision criterion reported separately rather than a single accuracy number. There is also a serious measurement floor to respect: human round-trip transliteration character error for Bengali runs 3.5–8.5%, and BenHalluEvals own conversions preserved meaning in 92.9% of cases, which is the same order of magnitude as the effect being measured.

Contribution. If judges are script-biased, every multilingual leaderboard that scores transliterated or code-mixed output is reporting a number partly produced by its own instrument. That is a methodological result well beyond Bengali. Given the design load already carried by Direction 1, this is best treated as the second paper.

5.4 How they compose, and what to cut

Direction 1 requires the corpus. Once it exists, Direction 2 needs only a training-regime comparison on top of it, and Direction 3 needs an additional script arm. Running all three from one annotation campaign is far cheaper than running any two separately.

The clear advice from adversarial review, however, is that Direction 1 plus Direction 2 is already a full paper, and that adding Direction 3 at the same time produces a design that is underpowered on every axis. The recommendation is therefore to build the corpus with Direction 1 and 2 as the paper, design the corpus so the script arm can be added later without rebuilding it, and defer Direction 3 to a follow-up ; citing it explicitly as deferred work so reviewers see the omission is deliberate.

6. The consolidated proposal

Working name: Bangla-RAG (B-RAG). Alternatives are BanglaRAGTruth, which signals lineage, and BhrantiBench, from ভ্রান্তি, "false belief". This is decision D4.

6.1 The claim the paper is organised around

Adversarial review was unanimous on one point: "a multidimensional pipeline benchmark" is a scope statement, not a claim, and reviewers reject papers that measure many things while accepting papers that show one thing and say how it could have been falsified. The proposed spine is therefore a single sentence:

In Bangla retrieval-augmented generation, faithfulness degrades as a measurable function of evidence quality, and the residual failures are generator-side integration errors rather than retrieval errors ; established with ground-truth attribution baked

into the data collection, not inferred by a judge whose Bangla accuracy is itself unknown.

Everything else in the design exists to support, qualify or falsify that sentence.

6.2 Objective

Build and release the first Bangla corpus that combines naturally generated model output, real retrieved evidence of controlled quality, and human span-level faithfulness annotation. Use it to measure how evidence quality shapes hallucination in Bangla RAG, to establish against human judgement how far automatic evaluation can be trusted in Bangla, and to train and release the first Bangla hallucination detectors together with the scaling curve that tells other low-resource communities how much annotation they need to buy.

6.3 Research questions and hypotheses

	Research question	Hypothesis	What would falsify it
RQ1	How does faithfulness vary as retrieved evidence degrades from oracle to none?	Faithfulness falls monotonically with evidence quality, and the curves shape differs by model family	A flat curve, or one dominated by noise, would mean evidence quality is not the driver
RQ2	How much unfaithfulness is retrieval-induced versus generator-intrinsic?	A substantial share persists under oracle evidence, making it generator-intrinsic; the precedent is a Bangla agricultural system that retained a 4.05–7.00% hallucination floor even with oracle evidence	If almost nothing survives oracle evidence, the problem is retrieval and the papers framing must change
RQ3	How reliable is LLM-as-a-judge faithfulness evaluation in Bangla?	Markedly less reliable than the 68.4–76.8 balanced accuracy MEMERAG reports for English through Hindi; anchoring with annotated examples recovers part of the gap	A zero-shot judge already agreeing with native annotators at English-level accuracy
RQ4	How many native annotated examples does a Bangla detector actually need?	Native supervision beats English-only and machine-translated training, with returns flattening in the low hundreds of examples	A flat curve ; meaning translated data suffices, which is publishable as a negative result and changes the advice we give
RQ5	Do models abstain when no relevant evidence exists?	Bangla-capable models over-answer, mirroring NoMIRACLs finding that some models exceed 88% hallucination on non-relevant subsets	Models abstaining appropriately would be a genuinely good-news result worth reporting

6.4 Methodology architecture

Seven stages. A to E build the corpus; F measures three things with it; G releases everything. A second track supplies existing and machine-made labelled data, used only as training arms and never for evaluation.

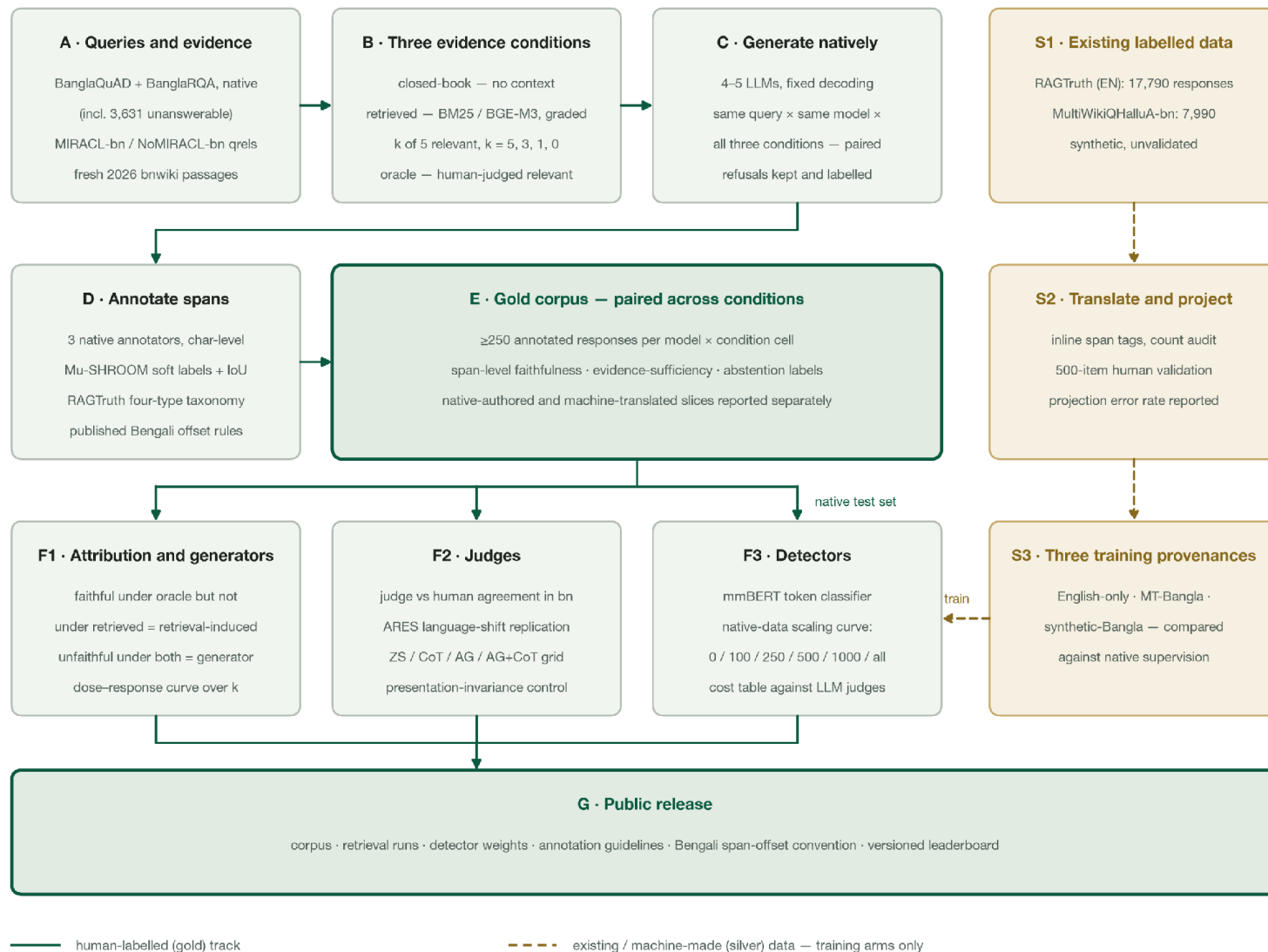


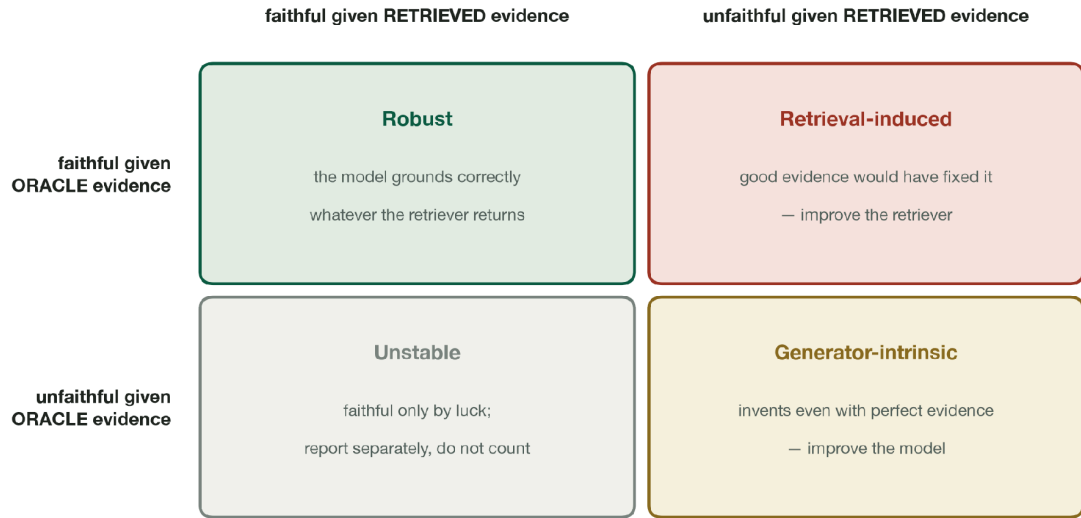
Figure 1. Proposed methodology architecture. The gold track (solid) is natively generated and human-annotated and produces the evaluation corpus. The silver track (dashed) ; English RAGTruth, its machine translation, and the existing synthetic Bengali split ; is used only to train detectors, never to test them.

Stage	What happens	Output	Drawn from
A · Queries and evidence	Native Bengali questions from BanglaQuAD (30,808 native-authored pairs) and BanglaRQA (14,889 pairs including 3,631 unanswerable). Passages from MIRACL-bn (297,265 passages, Apache-2.0) and NoMIRACL-bn, supplemented by a freshly chunked 2026 Bengali Wikipedia slice	Query sets and a passage index, with human relevance judgements attached	MEMERAG; deliberately avoids the sources BenHalluEval has consumed
B · Evidence conditions	Each query is presented under four conditions: closed-book; degraded retrieval (1 of 5 passages relevant); realistic retrieval (BM25 with the Lucene Bengali analyser, and BGE-M3 dense); and oracle (all passages human-judged relevant). A separate no-evidence arm uses NoMIRACLs human-verified all-non-relevant Bengali queries	Contexts of known, graded evidence sufficiency	NoMIRACL, extended from two buckets into a graded axis
C · Generate	Four to five LLMs answer in Bangla, fixed decoding, the same query answered by the same model under every condition ; a paired design	Naturally generated responses, refusals kept and labelled	RAGTruths natural elicitation
D · Annotate	Three native annotators mark hallucinated spans at character level under the Mu-SHROOM protocol, with soft probabilistic labels and IoU scoring binarised above 50% annotator agreement; plus evidence-sufficiency and abstention-correctness labels	Span labels with types, and the two relevance judgements	Mu-SHROOM scoring; RAGTruths four-type taxonomy; MEMERAGs flow-chart procedure
E · Corpus	Assemble, paired across conditions, and freeze. Native-authored and machine-translated slices reported separately	The benchmark	FaithJudges one-pool-many-uses logic
F1 · Attribution	The dose–response curve over evidence quality; attribution by construction (Figure 2); hallucination rate, span density and type distribution per model and condition; abstention and false-refusal rates	Generator leaderboard with diagnosis	New; the English diagnostic literature releases no corpus
F2 · Judges	Judge-versus-human agreement in Bangla; a replication of ARESs language-shift experiment; the full zero-shot / CoT / guidelines / guidelines+CoT grid; a presentation-invariance control	Judge reliability report and a usable Bangla judge	MEMERAG, FaithJudge, ARES
F3 · Detectors	mmBERT token classifiers trained under four data provenances, with a native-data scaling curve and a cost table against LLM judges	Open detector weights and the scaling curve	LettuceDetect, Turk-LettuceDetect, PsiloQA
G · Release	Corpus, retrieval runs, detector weights, guidelines, the Bengali span-offset convention, and a versioned leaderboard ; one permissive licence, working URLs from day one	Public artefacts	LettuceDetects release practice, explicitly not Turk-LettuceDetects

6.5 Attribution by construction

The paired design is what makes attribution ground truth rather than inference. A response that is faithful given oracle evidence but unfaithful given retrieved evidence is a retrieval-induced failure; one that is unfaithful under both is generator-intrinsic. This matters because if a Bangla judge is roughly 70–75% accurate on each of context relevance and faithfulness, the joint accuracy of a *judge-inferred* attribution is around 50% ; the central claim would sit at chance and the paper would not know it.

Same query, same generator, three evidence conditions



The third condition, closed-book, runs alongside as a control: it shows what the model would have said with no evidence at all, separating parametric recall from genuine grounding.

Attribution is fixed by the design, not inferred by a judge whose Bangla accuracy is itself unknown.

Figure 2. Attribution follows from the design. Because the same query is answered by the same model under oracle, retrieved and closed-book conditions, each failure is classified by comparing a model against itself rather than by asking a judge to diagnose it.

Two headline metrics fall out: the grounding-failure rate among cases where evidence was sufficient, which isolates the generator, and the abstention rate among cases where it was not, which measures whether a model knows when to stop.

6.6 What we build

Element	Specification	Rationale
Task 1 ; grounded QA	250 native Bengali queries × 4 generators × 4 evidence conditions = 4,000 responses	250 per model-condition cell, matching MEMERAGs per-language annotated volume, which yielded confidence intervals of ±2.0 to ±2.8 points
Abstention arm	150 NoMIRACL-bn all-non-relevant queries × 4 generators = 600 responses	Converts NoMIRACLs binary probe into free-text scoring; BanglaRQAs unanswerable questions supply a second source
Task 2 ; grounded summarisation	150 Bengali news sources × 4 generators × 2 conditions = 1,200 responses, budget permitting	A second task guards against single-task conclusions; it is the first thing to cut
Data-to-text	Dropped from v1.0	There is no native Bangla structured-record source, and RAGTruths data-to-text is a Yelp JSON artifact ; 68.6% of responses hallucinated, with about 17.7% of its spans caused by null values. Machine-translating it would import an English dataset artifact as a Bengali finding
Generators	4–5, chosen for generation competence: two frontier closed models, two open multilingual, and at most one Bangla-centric model, screened in a pilot	BenHalluEval found TituLLM and BanglaLLaMA degenerate; selecting for Bangla branding over competence would make several cells unusable

Element	Specification	Rationale
Script arm	Designed for, deferred to v1.1	Section 5.3; the corpus is built so it can be added without rebuilding
Pilot	30 queries × 4 generators × 4 conditions = 480 responses	A gate, not a formality

If annotation capacity is lower than planned, the fallback is 150 queries and three conditions ; 1,800 responses which roughly doubles confidence-interval width and should be accompanied by an explicit statement of the smallest effect the study can then detect.

6.7 Annotation protocol

- Guidelines. RAGTruths four-type taxonomy, written bilingually, extended with Bengali-specific categories no existing resource covers: honorific and register shift, Bengali numeral and date handling, transliterated named-entity spelling variants, and Bengali-versus-Bangladeshi entity ambiguity.
- A published span-offset convention. Bengali character offsets are not well defined without one: conjunct ligatures, two-part dependent vowel signs, nukta variants and ZWJ/ZWNJ all change IoU depending on normalisation. We fix NFC, state the rules, and publish them. None of Mu-SHROOMs fourteen languages had to resolve this, so it is a small genuine contribution.
- Procedure. MEMERAGs flow-chart decision process rather than free-form judgement ; the one documented intervention that moved agreement from Gwets AC1 0.45 to 0.84–0.93 ; with LLM-highlighted candidate evidence as an annotator aid.
- Pilot gate. Three annotators on 480 responses; iterate until coarse agreement clears 0.80. Full annotation does not start until it does.
- Main pass. Three annotators per response with soft labels, following Mu-SHROOM rather than a two-annotator-plus-adjudication scheme, so that annotator uncertainty is preserved rather than resolved away.
- Reporting. Gwets AC1, Fleiss κ and character-level Krippendorffs α , plus judge-versus-human concordance ; the number BenHalluEval never reports. RAGTruths 78.8% span-level consistency and MEMERAGs fine-grained collapse to AC1 0.18 are the realistic reference points; a modest κ honestly reported is defensible, an unreported one is not.

6.8 Evaluation protocol

Layer	What is scored	Metrics	Statistical treatment
F1 · Generators and attribution	4–5 LLMs × 4 evidence conditions	Hallucination rate, spans per 100 words, type mix; grounding-failure and abstention rates; dose–response over evidence quality	Bootstrap confidence intervals; crossed random effects for query and generator, since responses are nested in queries and crossed with models; permutation tests at $\alpha = 0.05$; Holm correction across the comparison grid
F2 · Judges	LLM judges × prompting strategy × anchoring; plus RAGAS-bn and NLI baselines	Balanced accuracy and macro-F1 against human labels, reported per error type; verdict-format compliance; presentation-flip rate against the published 10.7–14.4% baseline	Prediction-powered confidence intervals; judge prompt language held constant; faithfulness and language compliance never scored in the same call
F3 · Detectors	mmBERT-base and small, Granite R2, mGTE, BanglaBERT as the 512-token floor	Character-level IoU and Spearman ρ over soft labels (Mu-SHROOM-compatible), plus response and span P/R/F1 (RAGTruth-compatible); throughput, latency and cost per thousand judgments	Native and translated test sets reported separately; truncation rates reported for every 512-token model, with a length-matched slice

Layer	What is scored	Metrics	Statistical treatment
Ground truth	The annotation	Gwets AC1, Fleiss K, character-level Krippendorffs α	Three annotators throughout; soft labels released

Realistic targets, from comparable work. A character-level IoU of 0.45–0.65 would be a credible first Bengali result, benchmarked against PsiloQAs 48.81–76.61 range for mmBERT-base across languages. Beating a frontier LLM judge by roughly ten points is the demonstrated pattern ; LettuceDetects multilingual models beat GPT-4.1-mini by 10.76 to 17.04 F1. Turk-LettuceDetects 0.7266 token F1 is the number to be compared against, on native rather than translated data.

6.9 Baselines

12. Prompted frontier-LLM span detection, with RAGTruth-style JSON prompts adapted to Bangla.
13. The RAGAS faithfulness pipeline with multilingual components ; the first test of whether the standard industry tool means anything in Bangla. Note that RAGAS *context relevance* is explicitly not used as the retrieval-side signal: it measures focus and redundancy, agreed with humans on only 0.70 of pairwise comparisons in English, and ARES beats it by roughly 59 points. Retrieval quality is measured with true IR metrics against MIRACLs judgements instead.
14. A multilingual NLI detector, zero-shot, testing whether existing multilingual claims hold for Bengali.
15. English LettuceDetect via translate-test.
16. mmBERT trained on machine-translated RAGTruth ; the Turk-LettuceDetect recipe, reproduced with the human validation it omitted.
17. mmBERT trained on the existing synthetic Bengali split (MultiWikiQHallaA-bn, 7,990 rows), which converts the closest competing artifact into an ablation arm.
18. mmBERT trained on native Bangla annotation, at each point of the scaling curve.
19. A FaithJudge-style anchored judge using our own annotated pool.

The comparison answering RQ4 is between 5, 6 and 7 on the native test set. The LLM-judge comparator must use MEMERAGs best configuration rather than zero-shot prompting, or the efficiency claim is won against a strawman.

6.10 Expected contributions

20. The first Bangla corpus combining natural generation, controlled real retrieval, and human span-level faithfulness annotation ; the three properties no existing Bengali resource has together.
21. The first dose–response measurement of faithfulness against evidence quality, in any language, using per-passage human relevance judgements.
22. Retrieval-induced versus generator-intrinsic attribution operationalised as an annotated resource rather than an English-only diagnostic method.
23. The first Bangla hallucination detectors, released openly, with a native-data scaling curve that quantifies what annotation buys.
24. The first human-anchored measurement of LLM-judge reliability in Bangla, including a replication of ARESs language-shift result.
25. A published Bengali character-level span-offset convention, reusable by any future Indic span-annotation effort.
26. A justified-abstention track distinguishing faithful answers, hallucinations, false refusals and true refusals.

6.11 Effort, timeline and risk

Annotation dominates. At roughly seven minutes per response and two annotators per response, plus a three-annotator subset and about 15% adjudication, the 4,000-response core corpus costs approximately 950 to 1,100 annotator-hours ; four annotators at 20 hours a week for around three months. Generation

and judging cost low thousands of dollars at frontier API prices; translating RAGTruth takes about twelve GPU-hours; each detector trains in about two hours on a single A100.

Risk	Level	Mitigation
Statistical power ; the most likely rejection reason	High	250 responses per model-condition cell rather than ~112; four generators rather than eight; pre-register the smallest effect of interest and analyse with crossed random effects rather than condition means; drop any inferential claim whose unit of analysis is the model
The headline manipulation returns a null	High	Bengali retrieval is strong ; BM25 reaches nDCG@10 0.508 and R@100 0.909 on MIRACL Bengali, and lexical beats dense ; so a naive BM25-versus-dense contrast would likely find nothing. This is exactly why the manipulation is <i>graded evidence contamination</i> with oracle and closed-book anchors, which cannot be swamped: the k = 0 and oracle arms differ by construction
Corpus staleness manufactures false retrieval failures	High	MIRACL Bengali freezes the February 2019 Wikipedia dump ; 63,762 articles against today's 190,685, about 33% coverage. Either chunk a fresh 2026 dump and re-judge a sample, or restrict to time-invariant questions; report the coverage figure either way as a validity control
Span-level agreement fails to reach a publishable floor	High	The pilot is a gate; MEMERAGs flow-chart procedure is the documented remedy; coarse labels carry headline results with fine-grained labels reported descriptively
Judge unreliability contaminates the analysis	Medium	Attribution is by construction, not judge inference; judge quality is measured as a result rather than assumed
Contamination and source collision	Medium	BenHalluEval has consumed TyDiQA-GoldP Bengali, BanglaCHQ-Summ and SOMADHAN, so we use BanglaQuAD and BanglaRQA instead. Qwen3-Embedding trains on MIRACL and Mr. TyDi, so it cannot serve as a clean retriever or judge on MIRACL-bn. RAGTruth has been public since January 2024 and frontier judges have plausibly seen it
Language and script drift rewards the wrong behaviour	Medium	Most open models show very high register defection to English on Romanised Indic input. Score script and language compliance separately, in a separate judge call, and validate the language identifier on our own data rather than trusting off-the-shelf tools
Scooping	Medium	Two active groups are close: one has published three adjacent Bengali papers in nine months and has signalled extending its design; another moved from citation-grounded Bengali generation to Bengali retrieval evaluation in six weeks. Pilot immediately, preprint at corpus completion, and check the live Kaggle Bengali hallucination competition before submitting
Licensing	Low	MIRACL, Mr. TyDi and NoMIRACL are Apache-2.0 and form a clean substrate; BenHalluEval is CC-BY-NC-SA-4.0, which constrains redistributing derivatives

7. Decisions to finalise

	Decision	Options	Recommendation
D1	Scope of v1.0	Directions 1+2; all three; Direction 1 alone	1+2. Review was unanimous that all three at once produces a design underpowered on every axis
D2	Corpus scale	4,000-response core; 1,800 fallback; larger	Run the 480-response pilot now; commit to 4,000 only if agreement clears the gate
D3	Second task	Include summarisation; QA only	QA only for v1.0 unless annotation capacity exceeds plan; data-to-text is dropped either way
D4	Name	Bangla-RAG / B-RAG; BanglaRAGTruth; BhrantiBench	Your call ; B-RAG signals scope, BanglaRAGTruth signals lineage, BhrantiBench signals identity
D5	Passage corpus	MIRACL-bn as-is (33% coverage of current Wikipedia); fresh 2026 dump	Fresh dump if we can afford re-judging, otherwise time-invariant queries ; but report the coverage figure regardless

	Decision	Options	Recommendation
		with re-judging; time-invariant queries only	
D6	Annotation workforce	University students; professional vendor; hybrid	Students with a rigorous protocol, following BenHalluEvals precedent; a vendor is the escalation path
D7	Generator roster	Which four or five; whether to include a Bangla-centric model at all	Screen candidates in the pilot and exclude degenerate responders, whatever their branding
D8	Venue and timing	Main-conference resource track; LoResLM; BLP workshop; a *SHROOM-style shared task	Preprint at corpus completion for priority. The *SHROOM organisers are natural collaborators, since Bengali entered their task only as a zero-shot afterthought

8. References

8.1 The eight papers reviewed

27. Niu, C., Wu, Y., Zhu, J., et al. 2024. *RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models*. ACL 2024, 10862–10878. aclanthology.org/2024.acl-long.585
28. Es, S., James, J., Espinosa-Anke, L., Schockaert, S. 2024. *RAGAs: Automated Evaluation of Retrieval Augmented Generation*. EACL 2024 Demonstrations, 150–158. aclanthology.org/2024.eacl-demo.16
29. Saad-Falcon, J., Khattab, O., Potts, C., Zaharia, M. 2024. *ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems*. NAACL 2024, 338–354. aclanthology.org/2024.naacl-long.20
30. Cruz Blandón, M. A., Talur, J., Charron, B., Liu, D., Mansour, S., Federico, M. 2025. *MEMERAG: A Multilingual End-to-End Meta-Evaluation Benchmark for Retrieval Augmented Generation*. ACL 2025, 22577–22595. aclanthology.org/2025.acl-long.1101
31. Kovács, Á., Recski, G. 2025. *LettuceDetect: A Hallucination Detection Framework for RAG Applications*. arXiv:2502.17125
32. Taş, S., El Huseyni, M., Ezerceci, Ö., Bayraktar, R., Terzioğlu, F. B. 2025. *Turk-LettuceDetect: Hallucination Detection Models for Turkish RAG Applications*. arXiv:2509.17671
33. Tamber, M. S., Bao, F. S., Xu, C., et al. 2025. *Benchmarking LLM Faithfulness in RAG with Evolving Leaderboards*. EMNLP 2025 Industry Track, 799–811. aclanthology.org/2025.emnlp-industry.54
34. Adib, S. E. S., Sani, A. A., Esham, E. A., Abrar, A., Tashdeed, I., Chowdhury, M. T. A. 2026. *BenHalluEval: A Multi-Task Hallucination Evaluation Framework for Large Language Models on Bengali*. arXiv:2605.31483

8.2 Work the proposal must be positioned against

35. Thoresen, R., Smart, A. 2026. *A Multilingual Hallucination Benchmark: MultiWikiQHALLU*. arXiv:2605.02504 ; a Bengali split of 7,990 span-labelled synthetic examples already exists at huggingface.co/datasets/alexandrinst/multi-wiki-qa-synthetic-hallucinations. Must be cited and dismantled: no human validation, gold context rather than retrieval, no Bengali results reported.
36. Thakur, N., Bonifacio, L., Zhang, X., et al. 2024. *"Knowing When You Don't Know": A Multilingual Relevance Assessment Dataset for Robust Retrieval-Augmented Generation (NoMIRACL)*. Findings of EMNLP 2024. arXiv:2312.11361 ; includes Bengali; the direct predecessor for retrieval-induced hallucination, extended here from a binary probe to free-text span scoring.
37. Thakur, N., Kazi, S., Luo, G., Lin, J., Ahmad, A. 2025. *MIRAGE-Bench: Automatic Multilingual Benchmark Arena for RAG Systems*. NAACL 2025. arXiv:2410.13716 ; Bengali among 18 languages, but oracle passages and a surrogate judge, with no per-instance faithfulness labels.

38. Elchafei, R., Swain, M., Masoudian, S., Schedl, M. 2026. *Facet-Level Tracing of Evidence Uncertainty and Hallucination in RAG*. arXiv:2604.09174 ; already separates retrieval-induced from generator-intrinsic hallucination in English, as a diagnostic method releasing no corpus.
39. *SHROOM-CAP: Shared Task on Hallucinations in Crosslingual Analyses of Publications*. CHOMPS 2025 @ IJCNLP-AAACL ; Bengali present only as a zero-shot, test-only language with binary labels.
40. Datta, A., Chilukuri, S., Kumar, A., Ghosh, S., Zafar, A. 2026. *Do LLM Hallucination Detectors Suffer from Low-Resource Effect?* arXiv:2601.16766 ; studies Bengali directly; finds detectors robust within a language but failing cross-lingually without in-language supervision. The strongest single motivation citation for native annotation.
41. *BHRAM-IL: A Hallucination Benchmark for Indian Languages*. arXiv:2512.01852 ; Hindi, Gujarati, Marathi, Odia; Bengali absent.
42. Reza, K. R. I., Maria, N., Nimi, S. T. 2026. *Where Does Retrieval Fail? Evaluating RAG Architectures for Agricultural Advisory*. arXiv:2608.14886 ; Bengali retrieval evaluation; explicitly does not evaluate generation faithfulness or separate retrieval from generation failure.
43. Reza, K. R. I., Nimi, S. T., Shahid, R. 2026. *KrishokChat: A Citation-Grounded Dataset and Benchmark for Bengali Agricultural Advisory*. arXiv:2606.29243 ; reports a persistent 4.05–7.00% hallucination floor under oracle evidence.
44. Rafid, R., Adib, S., Ahmed, F., Abrar, A., Islam, M. 2026. *BanglaSummEval*. LoResLM 2026, aclanthology.org/2026.loreslm-1.51 ; reference-free factual-consistency metric for Bangla summarisation; Pearson r 0.694.

8.3 Resources the proposal depends on

45. Zhang, X., Thakur, N., Ogundepo, O., et al. 2023. *MIRACL: A Multilingual Retrieval Dataset Covering 18 Diverse Languages*. TACL ; Bengali subset: 1,631 train and 411 dev queries with human judgements over 297,265 passages, Apache-2.0.
46. Rashad, M., Rahman, M., et al. 2024. *BanglaQuAD: A Bengali Open-Domain Question Answering Dataset*. LREC-COLING 2024. arXiv:2410.10229 ; 30,808 native-authored pairs; not consumed by BenHalluEval.
47. *BanglaRQA: A Benchmark Dataset for Under-Resourced Bangla Language Reading Comprehension-Based Question Answering*. Findings of EMNLP 2022 ; 14,889 pairs including 3,631 unanswerable questions, the seed for the abstention track.
48. Marone, M., Weller, O., Fleshman, W., Yang, E., Lawrie, D., Van Durme, B. 2025. *mmBERT: A Modern Multilingual Encoder with Annealed Language Learning*. arXiv:2509.06888 ; 140M and 307M parameters, 8,192-token context, MIT; Bengali enters at mid-training with 3.9B tokens.
49. *PsiloQA: Multilingual Span-Level Hallucination Detection*. arXiv:2510.04849 ; establishes mmBERT-base as the best encoder for multilingual span-level hallucination detection, at IoU 48.81–76.61.
50. Chen, J., Xiao, S., Zhang, P., et al. 2024. *M3-Embedding (BGE-M3)*. Findings of ACL 2024. arXiv:2402.03216 ; the dense retriever, with published Bengali MIRACL nDCG@10 of 80.0.
51. Yang, E., Zhang, X., Lin, J. 2022. *Better Than Whitespace: Information Retrieval for Languages without Custom Tokenizers*. arXiv:2210.05481 ; justifies the Lucene Bengali analyser: MRR@100 0.418 against 0.245 for WordPiece on Mr. TyDi Bengali.
52. Vázquez, R., et al. 2025. *SemEval-2025 Task 3: Mu-SHROOM*. aclanthology.org/2025.semeval-1.322 ; the character-level span annotation and IoU scoring protocol adopted here; its 14 languages do not include Bengali.
53. Bhattacharjee, A., Hasan, T., Ahmad, W. U., et al. 2022. *BanglaBERT*. Findings of NAACL 2022. arXiv:2101.00204 ; the 512-token monolingual floor; BLUB NLI 82.80.
54. Bao, F. S., et al. 2025. *FaithBench: A Diverse Hallucination Benchmark for Summarization by Modern LLMs*. NAACL 2025 ; the annotated pool behind FaithJudge.

55. *Bi'an: A Bilingual Benchmark and Model for Hallucination Detection in Retrieval-Augmented Generation*. 2025. arXiv:2502.19209 ; the Chinese/English precedent for a per-language RAG hallucination benchmark.